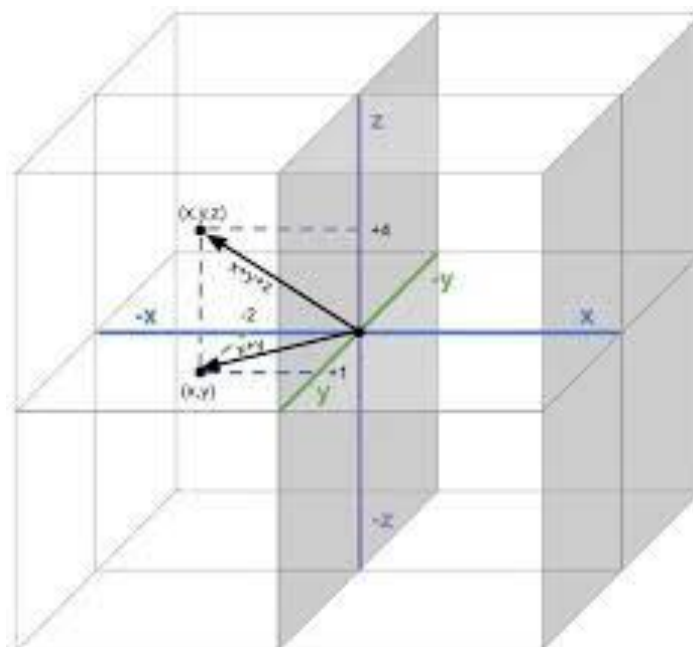


Dr. Mortaza Baky Haskuee



17 The Rank Theorem

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Lecture 17

The Rank Theorem

17.1 The Rank of a Matrix

We now give the definition to the rank of a matrix.

Definition 17.1: The **rank** of a matrix $\mathbf{A}$ is the dimension of its column space. We will use $\text{rank}(\mathbf{A})$ to denote the rank of $\mathbf{A}$.

Recall that $\text{Col}(\mathbf{A}) = \text{Range}(\mathbf{T}_{\mathbf{A}})$, and thus the rank of $\mathbf{A}$ is the dimension of the range of the linear mapping $\mathbf{T}_{\mathbf{A}}$. The range of a mapping is sometimes called the **image**.

We now define the nullity of a matrix.

Definition 17.2: The **nullity** of a matrix $\mathbf{A}$ is the dimension of its nullspace $\text{Null}(\mathbf{A})$. We will use $\text{nullity}(\mathbf{A})$ to denote the nullity of $\mathbf{A}$.

Recall that $\text{Null}(\mathbf{A}) = \ker(\mathbf{T}_{\mathbf{A}})$, and thus the nullity of $\mathbf{A}$ is the dimension of the kernel of the linear mapping $\mathbf{T}_{\mathbf{A}}$.

The rank and nullity of a matrix are connected via the following fundamental theorem known as the Rank Theorem.

Theorem 17.3: (Rank Theorem) Let $\mathbf{A}$ be a $m \times n$ matrix. The rank of $\mathbf{A}$ is the number of leading 1's in its RREF. Moreover, the following equation holds:

$$n = \text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}).$$

Proof. A basis for the column space is obtained by computing $\mathbf{rref}(\mathbf{A})$ and identifying the columns that contain a leading 1. Each column of $\mathbf{A}$ corresponding to a column of $\mathbf{rref}(\mathbf{A})$ with a leading 1 is a basis vector for the column space of $\mathbf{A}$. Therefore, if r is the number of leading 1's then $r = \text{rank}(\mathbf{A})$. Now let $d = n - r$. The number of free parameters in the

solution set of $\mathbf{A}\mathbf{x} = \mathbf{0}$ is d and therefore a basis for $\text{Null}(\mathbf{A})$ will contain d vectors, that is, $\text{nullity}(\mathbf{A}) = d$. Therefore,

$$\text{nullity}(\mathbf{A}) = n - \text{rank}(\mathbf{A}).$$

□

Example 17.4. Find the rank and nullity of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & 2 & 3 & -6 \\ 0 & -1 & -3 & 1 & 1 \\ -2 & 4 & -3 & -6 & 11 \end{bmatrix}.$$

Solution. Row reduce far enough to identify where the leading entries are:

$$\mathbf{A} \xrightarrow{2R_1+R_2} \begin{bmatrix} 1 & -2 & 2 & 3 & -6 \\ 0 & -1 & -3 & 1 & 1 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix}$$

There are $r = 3$ leading entries and therefore $\text{rank}(\mathbf{A}) = 3$. The nullity is therefore $\text{nullity}(\mathbf{A}) = 5 - \text{rank}(\mathbf{A}) = 2$. □

Example 17.5. Find the rank and nullity of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & -3 & -1 \\ -1 & 4 & 2 \\ -1 & 3 & 0 \end{bmatrix}.$$

Solution. Row reduce far enough to identify where the leading entries are:

$$\mathbf{A} \xrightarrow{R_1+R_2, R_1+R_3} \begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

There are $r = 3$ leading entries and therefore $\text{rank}(\mathbf{A}) = 3$. The nullity is therefore $\text{nullity}(\mathbf{A}) = 3 - \text{rank}(\mathbf{A}) = 0$. Another way to see that $\text{nullity}(\mathbf{A}) = 0$ is as follows. From the above computation, $\mathbf{A}$ is invertible. Therefore, there is only one vector in $\text{Null}(\mathbf{A}) = \{\mathbf{0}\}$. The subspace $\{\mathbf{0}\}$ has dimension zero. □

Using the rank and nullity of a matrix, we now provide further characterizations of invertible matrices.

Theorem 17.6: Let $\mathbf{A}$ be a $n \times n$ matrix. The following statements are equivalent:

- (i) The columns of $\mathbf{A}$ form a basis for $\mathbb{R}^n$.
- (ii) $\text{Col}(\mathbf{A}) = \mathbb{R}^n$
- (iii) $\text{rank}(\mathbf{A}) = n$
- (iv) $\text{Null}(\mathbf{A}) = \{\mathbf{0}\}$

- (v) $\text{nullity}(\mathbf{A}) = 0$
- (vi) $\mathbf{A}$ is an invertible matrix.

After this lecture you should know the following:

- what the rank of a matrix is and how to compute it
- what the nullity of a matrix is and how to compute it
- the Rank Theorem